

## 4.3 Interval Estimation

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### 4.3.1 Revisiting Sampling Distributions

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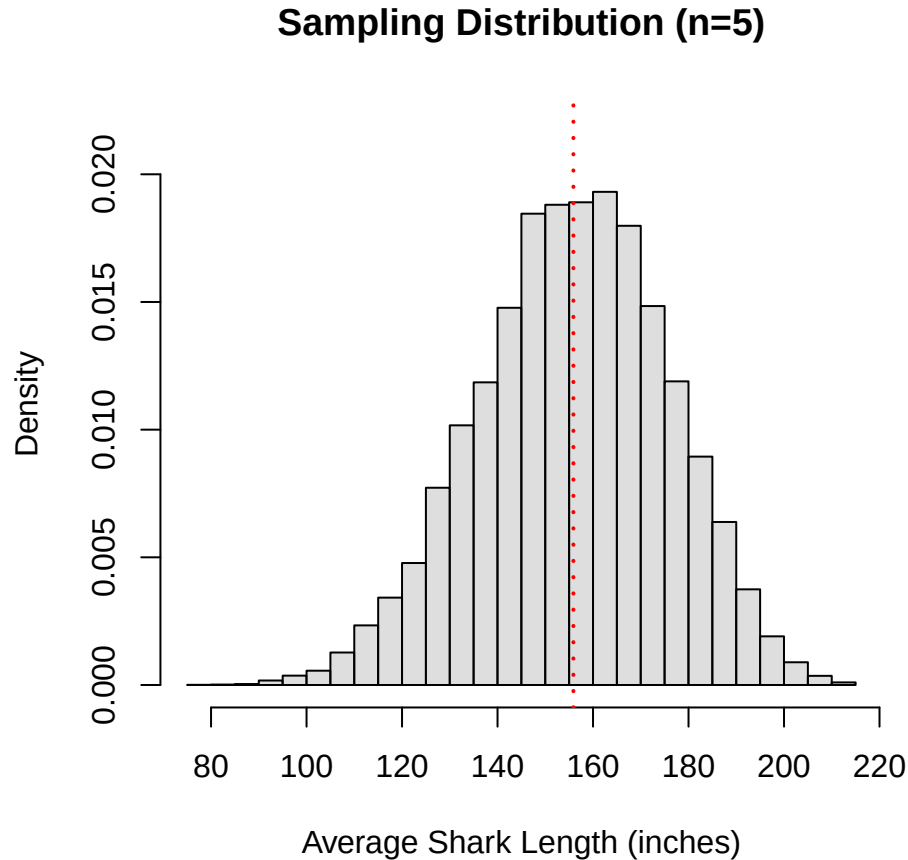
#### Overview

- As an introduction to Interval Estimation and Coverage Probabilities we revisit Sampling Distributions
- 

- Recall that when we look at  $a(\mathcal{S})$  for all possible samples  $\mathcal{S}$  of some size  $n$  from a population  $\mathcal{P}$  and that the values of  $a(\mathcal{S})$  have a distribution.
  - We called this the **sampling distribution** of  $\tilde{a}(\mathcal{S})$ .

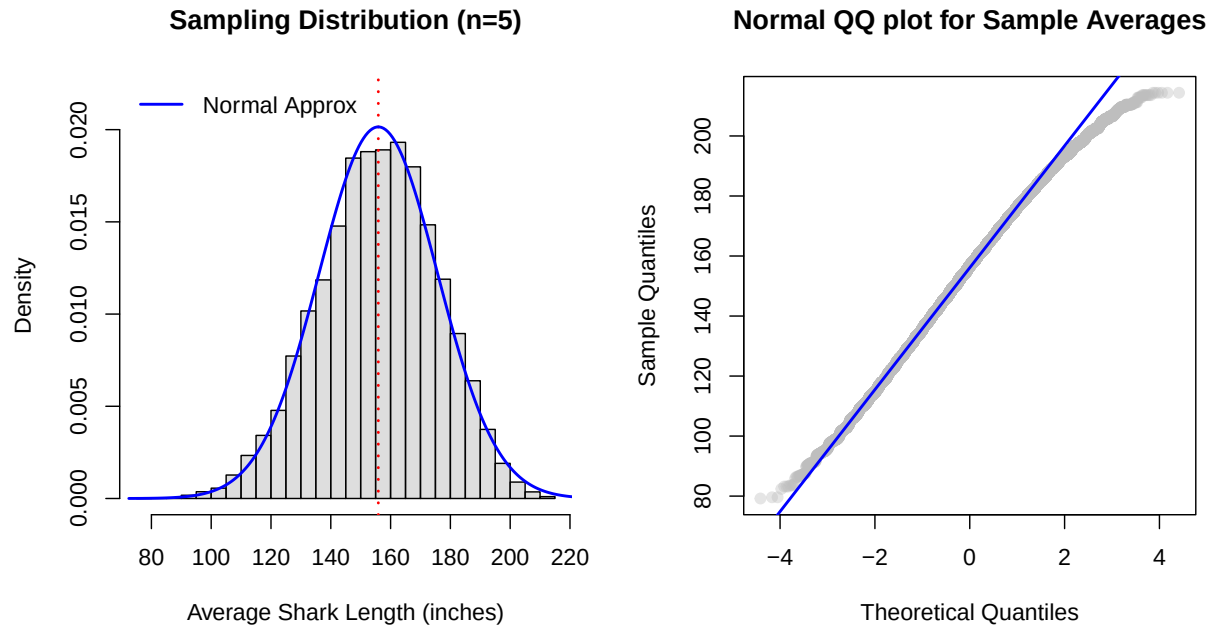
- For example, we looked at the Australian shark encounter population ( $N = 28$ ), and we considered all possible samples of size  $n = 5$ .

- We calculated  $a(\mathcal{S}) = \bar{y}$ , the average shark length, for each sample and then constructed a histogram.



- The histogram is fairly symmetric and centered about the population average  $a(\mathcal{P}) = 155.89$ 
  - The 2.5<sup>th</sup> and 97.5<sup>th</sup> quantiles of this distribution were 115.4 and 192.8
  - Thus we know that the interval (115.4, 192.8) contains the 95% of the possible sample averages.
- Quantifying the spread in this way is very useful – but in order to do so exactly we require having observed **all possible samples**.
  - Alternatively, we could instead consider a large number of possible samples to **approximate** the sampling distribution.
  - Or, we could approximate the sampling distribution via the **normal distribution** to simplify calculations.
  - Or, we could approximate the sampling distribution via **resampling techniques** such as the **bootstrap** (see Section 4.4).

- The normal distribution that best approximates the sampling distribution is the one with mean and standard deviation equal to the mean,  $\text{mean}(\text{avesSamp}) = 155.89$  and standard deviation,  $\text{sd}(\text{avesSamp}) = 19.8$ , from all possible  $a(S)$  values.
  - The plots below evaluate the adequacy of this approximation.



- The plots suggest that the normal approximation is pretty good, except for in the tails.
  - This is apparent in both the histogram and the QQ-plot.
  - The closeness of this approximation is unsurprising – recall the **central limit theorem (CLT)** and its implications for distributions of averages
- The normal approximation provides a model for the sampling distribution and can be used as a basis to construct confidence intervals for population averages.

## Summary

- The sampling distribution of some attributes can be approximated by a normal distribution.

## 4.3.2 Random vs. Observed Intervals

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### Overview

- We see how to construct a confidence interval, and we emphasize the distinction between random and observed intervals.
- 

- Suppose the attribute of interest is the population average  $a(\mathcal{P}) = \mu = \frac{1}{N} \sum_{u \in \mathcal{P}} y_u = \bar{y}$
- Recall (from STAT 231) that the estimator  $\tilde{a}(S) = \tilde{\mu} = \bar{Y}$  (a random variable) has the following properties:

$$E[\bar{Y}] = \mu \quad \text{and} \quad \text{Var}[\bar{Y}] = \frac{\sigma^2}{n}$$

where

$$\sigma^2 = \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \mu)^2$$

is the population variance.

- If the normality assumption holds (this may be appropriate due to the CLT) then the estimator  $\tilde{a}(\mathcal{P}) = \bar{Y}$  has the following distribution:

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- In the case of the Australian shark data

```
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)
popSharksAustralia <- popSharks[sharks$Australia == 1]

samples <- combn(popSharksAustralia, 5)
M <- ncol(samples)

avePop <- mean(sharks[popSharksAustralia, "Length"])
avesSamp <- apply(samples, MARGIN = 2, FUN = function(s) {
  mean(sharks[s, "Length"])
})

sdN = function(x) {
  sqrt(var(x) * (length(x) - 1)/length(x))
}
```

- Some attributes were
  - $N = 28$  and  $n = 5$
  - $\mu = a(\mathcal{P}) = 155.89$  is the population average
  - $\sigma = 47.97$  is the population standard deviation
  - $SD[\bar{Y}] = \frac{\sigma}{\sqrt{n}} = 19.8$ .
- Standardizing this random variable yields

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

- **Note:** Strictly speaking we are dealing with a finite population and so our variance term should include the **finite population correction**:

$$Var[\bar{Y}] = \frac{\sigma^2}{n} \left( \frac{N-n}{N-1} \right)$$

(Recall this was derived in the context of the variance of the Horvitz-Thompson estimator under SRSWOR)

- However, when  $N \gg n$  then  $\frac{N-n}{N-1} \rightarrow 1$  in which case this term can be ignored
- This is why most texts omit the correction factor, giving rise to the more familiar formulation  $Var[\bar{Y}] = \frac{\sigma^2}{n}$
- For completeness we will henceforth include the finite population correction, in which case

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n} \left( \frac{N-n}{N-1} \right)\right)$$

and

$$Z = \frac{\bar{Y} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} \sim N(0, 1)$$

- A histogram and corresponding  $N(0, 1)$  curve for this standardized quantity is visualized below:

```
N <- 28
n <- 5
fpc <- (N - n)/(n - 1)
Z = (avesSamp - mean(avesSamp))/sdN(avesSamp)

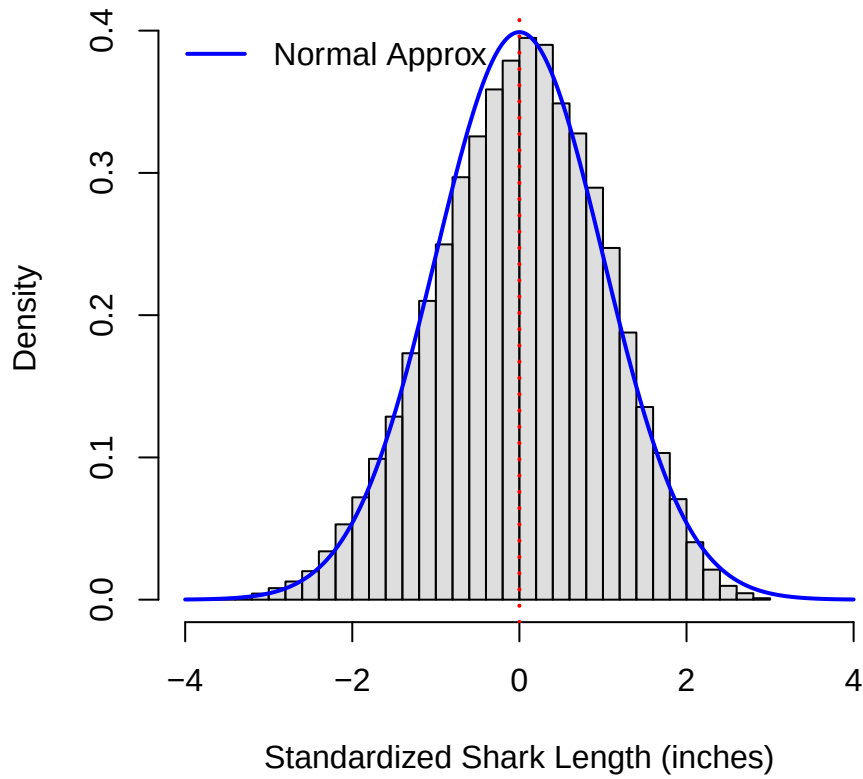
hist(Z, col = adjustcolor("grey", alpha = 0.5), freq = FALSE, main = "Sampling Distribution (n = 5)",
     xlab = "Standardized Shark Length (inches)", breaks = 25, xlim = c(-4, 4))

### Mark the population attribute in red
abline(v = 0, col = "red", lty = 3, lwd = 2)

### Add a Gaussian density
x = seq(-4, 4, length.out = 1000)
lines(x, dnorm(x), col = "blue", lwd = 2)

legend("topleft", legend = c("Normal Approx"), lwd = 2, col = "blue", lty = 1,
      bty = "n")
```

## Sampling Distribution (n = 5)



## Random Intervals

- Using the standardized random variable and specified  $p \in (0, 1)$  we can find a constant  $c > 0$  such that

$$\begin{aligned}
 1 - p &= Pr(-c \leq Z \leq c) \\
 1 - p &= Pr\left(-c \leq \frac{\bar{Y} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} \leq c\right) \\
 &= Pr\left(\bar{Y} - c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \leq \mu \leq \bar{Y} + c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}\right).
 \end{aligned}$$

- Rearranging this statement yields a **random interval** which contains  $\mu$  with probability  $1 - p$ .

$$\left[ \bar{Y} - c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{Y} + c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

- This interval is considered **random** because it's never actually observed – it is defined in terms of random variables
- However, observed intervals calculated by substituting  $\bar{Y}$  with  $\bar{y}$  are guaranteed to contain  $\mu$ ,  $100(1 - p)\%$  of the time.
  - $1 - p$  is therefore called the **coverage probability**.
  - $\mu$  is contained in (or *covered* by) such an interval  $100(1 - p)\%$  of the time.
  - All these intervals have the same width, they just have different (random) centres.

### A Note on determining $c$ :

- The normal distribution is symmetric about its mean  $\mu$ , so  $p$  and  $c$  are related through

$$1 - p = Pr(-c \leq Z \leq c)$$

or, equivalently,

$$1 - p/2 = Pr(Z \leq c)$$

where  $Z \sim N(0, 1)$  is a standard normal random variable.

- Therefore, given any  $p \in (0, 1)$  the value of  $c$  can be determined from the quantile function of a standard normal random variable:

$$c = Q_Z\left(1 - \frac{p}{2}\right)$$

which in R is calculated as `qnorm(1 - p/2)`.

- e.g.,  $c \approx 1.96$  when  $1 - p = 0.95$  for a standard normal random variable.

### Observed Intervals

- In practice, we will have only one sample
  - And thus a single numerical average  $\bar{y}$
  - And thus a single instance of these randomly generated intervals:

$$\left[ \bar{Y} - c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{Y} + c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

- In particular, we observe

$$\left[ \bar{y} - c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{y} + c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

- Such observed intervals are referred to as **confidence intervals** and you must take care to distinguish them conceptually from a random interval
  - In the context of random intervals, probabilistic statements are sensible.
  - However, an observed confidence interval is not random – it either contains  $\mu$  or it doesn't.
- Probability statements are in reference to the **method used to generate the intervals**, NOT to the particular interval we have observed.
- If the normality assumption holds up,  $100(1-p)\%$  of such intervals will contain  $\mu$ ,
  - We thus have some **confidence** that our particular observed interval will contain  $\mu$  as well, but unfortunately we'll never know if it does
- The larger  $1-p$ , the *more* confident we are that the interval will contain  $\mu$ .

## Example: Coverage Probability Estimation via Simulation

- Given that  $\mu = a(\mathcal{P}) = 155.89$  and  $SD(\bar{Y}) = \text{sdN}(\text{avesSamp}) = 19.8$ .
  - We randomly select 100 samples of size  $n = 5$
  - For each sample we calculate the 95% confidence interval (i.e.,  $(1-p) = 0.95$  and  $c \approx 1.96$ )
  - We should hope to find that approximately  $100(1-p)\% = 95\%$  of the intervals contain (i.e., cover)  $\mu = 155.89$ .

```
## R code to simulate the GAUSSIAN approximation
ylim <- c(0, 0.022)

### Add a Gaussian density
tmpAve <- mean(avesSamp)
tmpSD <- sdN(avesSamp)
confidence <- 0.95
p <- 1 - confidence
```



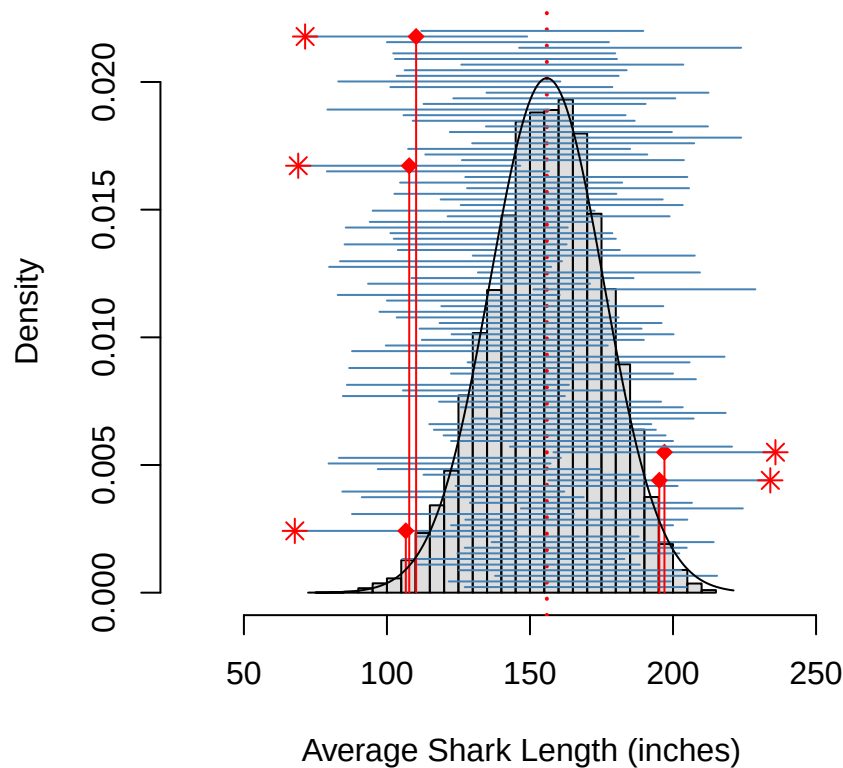
```

numIntervals <- 100
cValue <- qnorm(1 - p/2) # or qnorm((confidence + 1)/2)
set.seed(341) # comment out this line to get different samples every time you run the code.
ybarSampled <- sample(avesSamp, numIntervals)
heights <- seq(diff(ylim)/numIntervals, max(ylim), length.out = numIntervals)
xlim <- extendrange(avesSamp + cValue * c(-tmpSD, tmpSD))
hist(avesSamp, col = adjustcolor("grey", alpha = 0.5), freq = FALSE, main = paste0(numIntervals,
  " Individual ", round(100 * confidence), "% Confidence Intervals"), xlab = "Average Shark Length (in",
  ylim = ylim, xlim = xlim, breaks = 25)

### Mark the population attribute in red
abline(v = avePop, col = "red", lty = 3, lwd = 2)
numIntervalsMissed <- 0
for (i in 1:numIntervals) {
  lines(ybarSampled[i] + cValue * c(-tmpSD, tmpSD), rep(heights[i], 2), col = "steelblue")
  if (tmpAve > ybarSampled[i] + cValue * tmpSD) {
    points(ybarSampled[i] - cValue * tmpSD, heights[i], pch = 8, cex = 1.2,
      col = "red")
    points(ybarSampled[i], heights[i], pch = 18, cex = 1.2, col = "red")
    lines(rep(ybarSampled[i], 2), c(0, heights[i]), col = "red")
  } else if (tmpAve < ybarSampled[i] - cValue * tmpSD) {
    points(ybarSampled[i] + cValue * tmpSD, heights[i], pch = 8, cex = 1.2,
      col = "red")
    points(ybarSampled[i], heights[i], pch = 18, cex = 1.2, col = "red")
    lines(rep(ybarSampled[i], 2), c(0, heights[i]), col = "red")
  } else numIntervalsMissed <- numIntervalsMissed + 1
}
tmpX <- extendrange(avesSamp)
tmpX <- seq(tmpX[1], tmpX[2], length.out = 200)
lines(tmpX, dnorm(tmpX, mean = tmpAve, sd = tmpSD))

```

## 100 Individual 95% Confidence Intervals



- 95 of these 100 intervals cover the value  $\mu$  (which is indicated by the vertical dashed red line)
  - Those which *do not* cover  $\mu$  are marked with an asterisk and the corresponding sample means are indicated by diamonds and vertical solid red lines.
  - What do we notice?

## Summary

- We saw that a confidence interval is a realization of a random interval.
- Probability statements, such as “95% of the random intervals contain the true value”, are in reference to the method used to generate the intervals, NOT to the particular interval we have observed.
- The confidence intervals we calculated assumed  $SD[\bar{Y}]$  was known which is an unrealistic assumption.

### 4.3.3 Student $t$ Based Intervals

---

#### Overview

- We discuss how to construct a confidence interval using a  $t$ -distribution as an approximation to the sampling distribution.
- 

#### Standard Error vs. Standard Deviation

- In the previous section the confidence intervals we calculated assumed  $SD[\bar{Y}]$  was known
  - This is an unrealistic assumption.
  - Only very rarely would we have this value.
- However, for many sample attributes  $a(\mathcal{S})$  (e.g., Horvitz-Thompson estimators), we can estimate the standard deviation  $SD[\tilde{a}(\mathcal{S})]$  of the **sampling distribution** of  $\tilde{a}(\mathcal{S})$ .
  - The **standard error** is an estimate of the standard deviation of the corresponding estimator:

$$SE[\tilde{a}(\mathcal{S})] = \widehat{SD}[\tilde{a}(\mathcal{S})]$$

- We could use

$$\frac{a(\mathcal{S}) - a(\mathcal{P})}{SE[\tilde{a}(\mathcal{S})]}$$

- Note that using the estimate  $SE$  in place of  $SD$  will increase the variability of the random intervals.
- The corresponding estimator has much more variability – since we have to estimate  $\widehat{SD}$  now as well.

- **Example:** Let us illustrate this phenomenon using the Australia shark lengths data
  - Because we've taken all possible samples, we know  $SD[\tilde{a}(\mathcal{S})]$  but let's pretend like we don't and estimate it.
  - Thus let's calculate both

$$\frac{a(\mathcal{S}) - a(\mathcal{P})}{SD[\tilde{a}(\mathcal{S})]} \quad \text{and} \quad \frac{a(\mathcal{S}) - a(\mathcal{P})}{SE[\tilde{a}(\mathcal{S})]}$$

```
### Loading the shark data
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)
popSharksAustralia <- popSharks[sharks$Australia == 1]

samples <- combn(popSharksAustralia, 5)
M <- ncol(samples)

avePop <- mean(sharks[popSharksAustralia, "Length"])
avesSamp <- apply(samples, MARGIN = 2, FUN = function(s) {
  mean(sharks[s, "Length"])
})

sdN = function(x) {
  sqrt(var(x) * (length(x) - 1)/length(x))
}

Z = (avesSamp - mean(avesSamp))/sdN(avesSamp)

sdsSamp <- apply(samples, MARGIN = 2, FUN = function(s) {
  sdN(sharks[s, "Length"])
})

n = 5
N = 28
se = sdsSamp/sqrt(n) * sqrt((N - n)/(N - 1))
t = (avesSamp - mean(avesSamp))/se

## So the two histograms have the same bins
delta = 0.2
brk = seq(min(t) - delta, max(t) + delta, delta)

par(mfrow = c(1, 2))
hist(Z, col = adjustcolor("grey", alpha = 0.5), freq = FALSE, main = "Standardized averages \n with known",
     xlab = "Standardized Shark Length (inches)", breaks = brk, xlim = c(-4,
     4), ylim = c(0, 0.4))

### Mark the population attribute in red
abline(v = 0, col = "red", lty = 3, lwd = 2)

### Add a Gaussian density
x = seq(-4, 4, length.out = 1000)
lines(x, dnorm(x), lwd = 2, col = "blue")

hist(t, col = adjustcolor("grey", alpha = 0.5), freq = FALSE, main = "Standardized averages \n with estimated",
     xlab = "Standardized Shark Length (inches)", breaks = brk, xlim = c(-4,
```

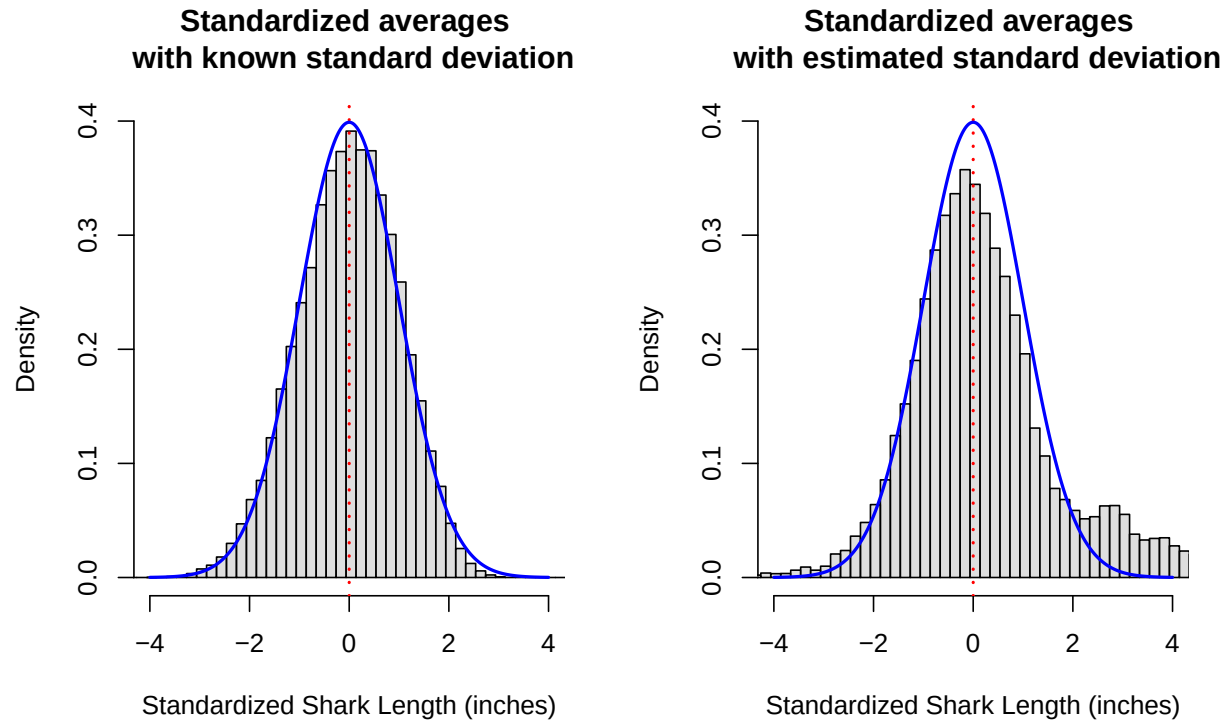
```

4), ylim = c(0, 0.4))

### Mark the population attribute in red
abline(v = 0, col = "red", lty = 3, lwd = 2)

### Add a Gaussian density
x = seq(-4, 4, length.out = 1000)
lines(x, dnorm(x), lwd = 2, col = "blue")

```



- The blue lines correspond to the  $N(0,1)$  density curve
  - the normal distribution does not seem to be a good approximation when we have to estimate the standard deviation.

- If the data are normally distributed we have the following distributional result

$$\frac{\bar{Y} - \mu}{\frac{\tilde{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} \sim t_{n-1}$$

- This statistic is called a **pivotal quantity** because
  - it is a function of the sample data  $(Y_u, u \in \mathcal{S})$  and unknown parameter  $(\mu)$  and its **sampling distribution is completely known**.

- Let's try the  $t_{n-1}$  distribution as an approximation instead of the  $N(0, 1)$ :

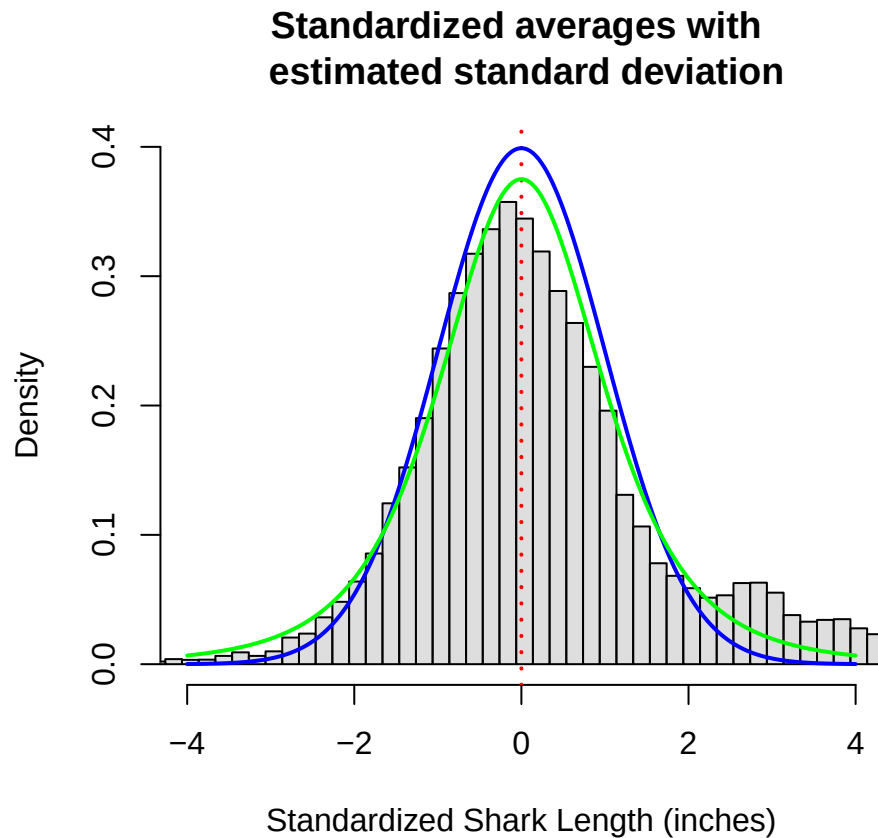
```
n = 5
N = 28
se = sdsSamp/sqrt(n) * sqrt((N - n)/(N - 1))
t = (avesSamp - mean(avesSamp))/se

## So the two histograms have the same bins
delta = 0.2
brk = seq(min(t) - delta, max(t) + delta, delta)

hist(t, col = adjustcolor("grey", alpha = 0.5), freq = FALSE, ylim = c(0, 0.4),
     main = "Standardized averages with \n estimated standard deviation", xlab = "Standardized Shark Length",
     breaks = brk, xlim = c(-4, 4))

### Mark the population attribute in red
abline(v = 0, col = "red", lty = 3, lwd = 2)

### Add a Gaussian density
x = seq(-4, 4, length.out = 1000)
lines(x, dnorm(x), col = "blue", lwd = 2)
lines(x, dt(x, n - 1), col = "green", lwd = 2)
```



- The green and blue curves represent the  $t$ -distribution and normal distribution density curves, respectively
  - The  $t$ -distribution with  $n - 1 = 4$  is a better approximation, but still not great.

- Now if we suppose that

$$\frac{\tilde{a}(\mathcal{S}) - a(\mathcal{P})}{\widetilde{SD}[\tilde{a}(\mathcal{S})]} \sim t_{n-1}$$

- Then we choose a  $p \in (0, 1)$  and a corresponding  $c > 0$  with

$$\begin{aligned} 1 - p &= Pr \left( -c \leq \frac{\tilde{a}(\mathcal{S}) - a(\mathcal{P})}{\widetilde{SD}[\tilde{a}(\mathcal{S})]} \leq c \right) \\ &= Pr \left( \tilde{a}(\mathcal{S}) - c \times \widetilde{SD}[\tilde{a}(\mathcal{S})] \leq \mu \leq \tilde{a}(\mathcal{S}) + c \times \widetilde{SD}[\tilde{a}(\mathcal{S})] \right) \end{aligned}$$

- *This* random interval has both a random center *and* a random length.
- The value of  $c$  is determined using the  $t$  distribution with  $n - 1$  degrees of freedom.
  - In R use `qt(1-p/2, df = n-1)` to get the value of  $c$
  - You can check that  $c \approx 2.78$  when  $1 - p = 0.95$  for a  $t_4$  random variable.

- In the special case that  $a(\mathcal{P}) = \bar{Y}$  is the population average then the standard deviation is

$$SD[\tilde{a}(\mathcal{S})] = \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N - n}{N - 1}}$$

and the standard deviation estimator is

$$\widetilde{SD}[\tilde{a}(\mathcal{S})] = \frac{\tilde{\sigma}}{\sqrt{n}} \times \sqrt{\frac{N - n}{N - 1}}$$

and the standard deviation estimate (the **standard error**) is

$$\widehat{SD}[\tilde{a}(\mathcal{S})] = \frac{\hat{\sigma}}{\sqrt{n}} \times \sqrt{\frac{N - n}{N - 1}}$$

- Thus the random interval in this case is

$$\left[ \bar{Y} - c \times \frac{\tilde{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{Y} + c \times \frac{\tilde{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right].$$

- And the corresponding observed confidence interval is

$$\left[ \bar{y} - c \times \frac{\hat{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{y} + c \times \frac{\hat{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right].$$

- **Note:** In the formulae above  $\sigma$  denotes the population standard deviation, which may be estimated in a sample by

$$\hat{\sigma} = \sqrt{\frac{\sum_{u \in \mathcal{S}} (y_u - \bar{y})^2}{n}}$$

## Example: Coverage Calculation via Simulation

- We randomly select 100 samples of size  $n = 5$ , and
- For each sample we calculate the 95% confidence interval (defined above)
  - using  $c \approx 2.78$  from the  $t_4$  distribution
- We should hope to find that approximately 95% of the intervals contain (or cover)  $\mu = 155.89$ .

```
## R code to simulate the t-STUDENT approximation
ylim <- c(0, 0.022)
n <- 5
N <- 28

### Add a Gaussian density
tmpAve <- mean(avesSamp)
tmpSD <- sdN(avesSamp)
confidence <- 0.95
numIntervals <- 100
cValue <- qt((confidence + 1)/2, df = n - 1)
set.seed(341)
tmpSamp <- sample(1:length(avesSamp), numIntervals)
ybarSampled <- avesSamp[tmpSamp]
sdSampled <- sdsSamp[tmpSamp]
maxSD <- max(sdSampled)/sqrt(n)

heights <- seq(diff(ylim)/numIntervals, max(ylim), length.out = numIntervals)
xlim <- extendrange(c(ybarSampled - cValue * sqrt((N - n)/(N - 1)) * sdSampled/sqrt(n),
  ybarSampled + cValue * sqrt((N - n)/(N - 1)) * sdSampled/sqrt(n)))
hist(avesSamp, col = adjustcolor("grey", alpha = 0.5), freq = FALSE, main = paste0(numIntervals,
  " Individual ", round(100 * confidence), "% Confidence Intervals"), xlab = "Average shark length (i",
  ylim = ylim, xlim = xlim, breaks = 25)
### Mark the population attribute in red
```



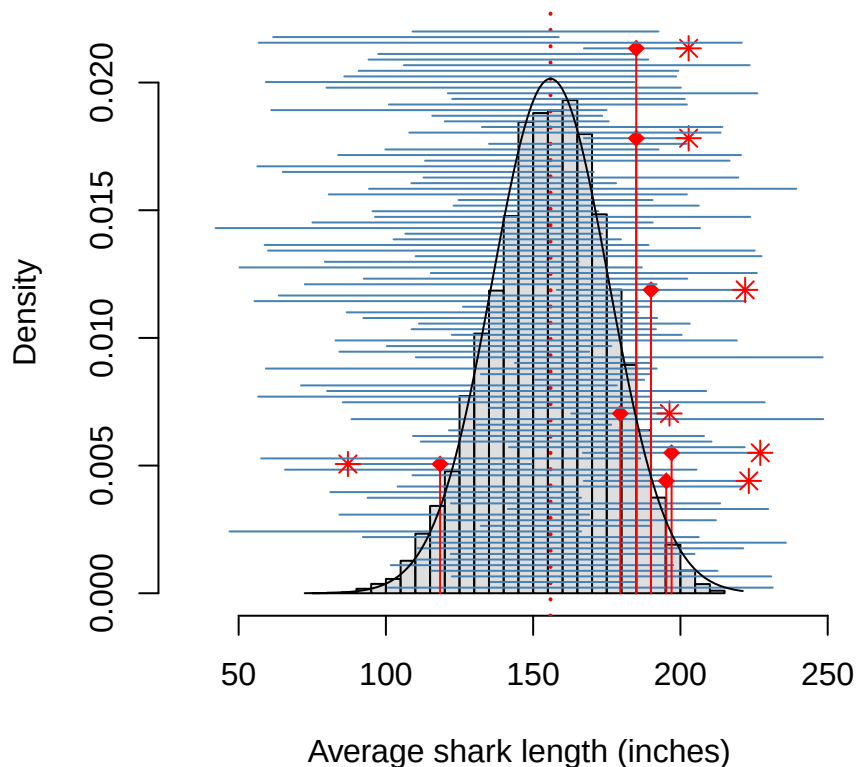
```

abline(v = avePop, col = "red", lty = 3, lwd = 2)
numIntervalsMissed <- 0
for (i in 1:numIntervals) {
  tmpSampSD <- sdSampled[i]/sqrt(n) * sqrt((N - n)/(N - 1))

  lines(ybarSampled[i] + cValue * c(-tmpSampSD, tmpSampSD), rep(heights[i],
    2), col = "steelblue")
  if (tmpAve > ybarSampled[i] + cValue * tmpSampSD) {
    points(ybarSampled[i] - cValue * tmpSampSD, heights[i], pch = 8, cex = 1.2,
      col = "red")
    points(ybarSampled[i], heights[i], pch = 18, cex = 1.2, col = "red")
    lines(rep(ybarSampled[i], 2), c(0, heights[i]), col = "red")
  } else if (tmpAve < ybarSampled[i] - cValue * tmpSampSD) {
    points(ybarSampled[i] + cValue * tmpSampSD, heights[i], pch = 8, cex = 1.2,
      col = "red")
    points(ybarSampled[i], heights[i], pch = 18, cex = 1.2, col = "red")
    lines(rep(ybarSampled[i], 2), c(0, heights[i]), col = "red")
  } else numIntervalsMissed <- numIntervalsMissed + 1
}
tmpX <- extendrange(avesSamp)
tmpX <- seq(tmpX[1], tmpX[2], length.out = 200)
lines(tmpX, dnorm(tmpX, mean = tmpAve, sd = tmpSD))

```

## 100 Individual 95% Confidence Intervals



- 93 of these 100 intervals cover the value  $\mu$  (which is indicated by the vertical dashed red line)
  - Those which *do not* cover  $\mu$  are marked with an asterisk and the corresponding sample means are indicated by diamonds and vertical solid red lines.
  - What do we notice?
- **Note** that we can find the exact coverage probability by calculating all possible confidence intervals.

### A Note on Pivotal Quantities:

- Pivotal quantities are the basis for constructing random intervals.
- Many random intervals are constructed via pivotal quantities such as

$$\frac{\tilde{a}(\mathcal{S}) - a(\mathcal{P})}{\widehat{SD}[\tilde{a}(\mathcal{S})]}$$

- The term *pivot* comes from the fact that with this quantity (which is a function of both  $\mathcal{S}$  and  $\mathcal{P}$ ) we are able to pivot and isolate for  $a(\mathcal{P})$
  - This is the general prescription for constructing random intervals.
- 
- This particular pivotal is a *t*-like pivotal, but it's not the *only* form of pivotal quantity.
    - One common pivotal quantity used for scale attributes  $s(\cdot)$ , are of the form

$$\frac{\tilde{s}(\mathcal{S})}{s(\mathcal{P})}$$

- e.g., this gives rise to  $\chi^2$ -based intervals for population variances.

So what have we learned?

## Summary

- We reviewed the notion of a pivotal quantity.
- We used a t-distribution as an approximation to the sampling distribution of the pivotal quantity to construct a confidence interval.